

Worksheet 5: Multiqubits and Quantum Circuits

Quantum Computation and Error Correction

Suggested time: 135 minutes — Reading: Nielsen & Chuang §1.3.1–1.3.6 and §4.3

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Conventions used throughout. Qubits are numbered from 0. A two-qubit state is written $|ab\rangle$, meaning qubit 0 is in state a and qubit 1 in state b ; matrices are written in the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.^a

The gate toolkit.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad |\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}.$$

- $\text{cx}(a, b)$ — flips qubit b if and only if qubit a is 1. Qubit a is the *control*, b the *target*.
- $\text{cz}(a, b)$ — multiplies the state by -1 if and only if both qubits are 1.
- $\text{swap}(a, b)$ — exchanges the states of qubits a and b .

Two rules for turning circuits into matrices. Gates applied to *different* qubits in the same time step combine as a tensor product, e.g. H on qubit 0 and X on qubit 1 is $H \otimes X$. Gates applied *one after another* combine as a matrix product in reverse order: the circuit “first G_1 , then G_2 ” is the matrix $G_2 G_1$. We write $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$ for the two projectors on a single qubit.

Timing guide: Part A $\approx$ 40 min Part B $\approx$ 50 min Part C $\approx$ 45 min.

^aQiskit numbers qubits the same way but prints bitstrings with qubit 0 on the *right*. Keep this in mind if you check anything on a computer.

Part A — Two qubits: states and gates as matrices

1. Tensor products in practice. A two-qubit state is a vector with four components.

- (a) Write $|01\rangle$ and $|+\rangle \otimes |-\rangle$ as four-component column vectors.
- (b) Write the general two-qubit state as $|\psi\rangle = \sum_{a,b \in \{0,1\}} c_{ab} |ab\rangle$. Show that $|\psi\rangle$ is a *product state*, i.e. can be written as $|\alpha\rangle \otimes |\beta\rangle$ for some single-qubit states $|\alpha\rangle, |\beta\rangle$, if and only if

$$c_{00} c_{11} = c_{01} c_{10}.$$

Hint: arrange the four amplitudes as a 2×2 matrix C with entries $C_{ab} = c_{ab}$. What does the product form say about C ?

- (c) Use (b) to decide which of the following are product states, and factorise the ones that are:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}, \quad \frac{|00\rangle + |01\rangle - |10\rangle - |11\rangle}{2}.$$

- (d) Using $(A \otimes B)(C \otimes D) = AC \otimes BD$, show that two gates acting on different qubits commute: applying H to qubit 0 and then X to qubit 1 gives the same matrix as doing it in the other order. This is why a circuit diagram does not need to say which of two gates in the same column is done “first”.

2. The two workhorses: cx and cz .

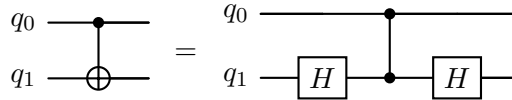
- (a) Write $\text{cx}(0, 1)$ and $\text{cx}(1, 0)$ as 4×4 matrices. Both are permutation matrices (exactly one 1 in each row and column). Explain in one sentence why that had to be the case.

- (b) Show that $\text{cx}(0, 1)$ can be written as

$$\text{cx}(0, 1) = P_0 \otimes \mathbb{I} + P_1 \otimes X,$$

and read this formula in words. Write the analogous formula for $\text{cx}(1, 0)$.

- (c) $\text{cz}(0, 1)$ is the diagonal matrix $\text{diag}(1, 1, 1, -1)$. Write it in the same projector form as (b), and show that $\text{cz}(0, 1) = \text{cz}(1, 0)$: for this gate it is meaningless to ask which qubit is the control. Is the same true of cx ?
- (d) Show that $\text{cx}(0, 1) = (\mathbb{I} \otimes H) \text{cz}(0, 1) (\mathbb{I} \otimes H)$, i.e. that the circuits

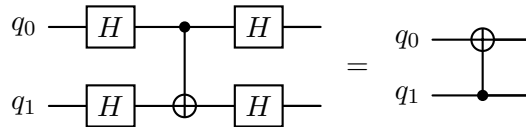


are the same. So cx and cz differ only by a change of basis on the target. *Hint: $HZH = X$.*

- (e) **Phase kick-back.** Compute $\text{cx}(0, 1) |+\rangle |0\rangle$ and $\text{cx}(0, 1) |+\rangle |-\rangle$. In the second case the target qubit is unchanged and it is the *control* that has changed. Explain how this is consistent with the description “flips qubit 1 if qubit 0 is 1”.

Part B — Circuit identities

3. **Hadamards turn a cx around.** Show that



that is, $(H \otimes H) \text{cx}(0, 1) (H \otimes H) = \text{cx}(1, 0)$. Do it in the following steps rather than by multiplying 4×4 matrices.

- (a) Verify $HXH = Z$ and $HZH = X$.
- (b) Using $P_0 = \frac{1}{2}(\mathbb{I} + Z)$ and $P_1 = \frac{1}{2}(\mathbb{I} - Z)$, expand the formula of Q2(b) into a sum of tensor products of Paulis:

$$\text{cx}(0, 1) = \frac{1}{2}(\mathbb{I} \otimes \mathbb{I} + Z \otimes \mathbb{I} + \mathbb{I} \otimes X - Z \otimes X).$$

- (c) Conjugate every term by $H \otimes H$ and regroup the result into the projector form of $\text{cx}(1, 0)$.
- (d) In the computational (Z) basis, cx acts as $|ab\rangle \mapsto |a, a \oplus b\rangle$ with $a, b \in \{0, 1\}$: the control is unchanged and its value is added (mod 2) to the target. Find the basis in which the roles are reversed, i.e. in which the target is unchanged and its value is added to the control.

4. **Three cx gates make a swap.** Worksheet 1 stated that a **swap** costs three cx gates. Now prove it.

- (a) Follow a computational basis state $|ab\rangle$ through the circuit $\text{cx}(0, 1)$, $\text{cx}(1, 0)$, $\text{cx}(0, 1)$ and show that it comes out as $|ba\rangle$. *Hint: keep track of the two bits using $\oplus$, and use $a \oplus a = 0$.*
- (b) You checked only the four basis states. Why is that enough to conclude that the circuit swaps *every* two-qubit state, entangled ones included?
- (c) Show that no circuit made of just *two* cx gates (in whichever directions) is a **swap**. *Hint: there are only four such circuits, and it is enough to follow $|01\rangle$ and $|10\rangle$ through each.*

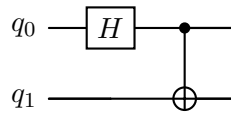
5. Pushing errors through a cx. Suppose a stray X (a bit flip) or Z (a phase flip) hits one qubit just *before* a cx . Where has it gone *after* the gate? Since $\text{cx}^2 = \mathbb{I}$, the question is what $\text{cx}(P)\text{cx}$ is for a Pauli P on one of the qubits.

- Show that $\text{cx}(0, 1)(X \otimes \mathbb{I})\text{cx}(0, 1) = X \otimes X$: a bit flip on the control is *copied* onto the target. *Hint: use the projector form of Q2(b) and $XP_0X = P_1$.*
- Show that $\text{cx}(0, 1)(\mathbb{I} \otimes Z)\text{cx}(0, 1) = Z \otimes Z$: a phase flip on the target is copied onto the control.
- Show that X on the target and Z on the control pass through unchanged.
- A single X error hits qubit 0 at the start of the circuit $\text{cx}(0, 1)$, $\text{cx}(1, 2)$ acting on three qubits. Use (a) and (c) to show that the circuit turns it into $X \otimes X \otimes X$: one bit flip has become three. Draw the circuit.

This “error tracking” is exactly how syndromes are computed in quantum error correction, which you will meet later in the course.

Part C — Entanglement from circuits

6. The Bell circuit. Consider



- Compute the output for each of the four inputs $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. These four outputs are the *Bell states*; check that they are orthonormal, and use Q1(b) to confirm that none of them is a product state.
- Every circuit can be run backwards. Write down the circuit that maps each Bell state back to the computational basis state it came from. (This is how a *Bell measurement* is done in practice: run this circuit, then measure both qubits in the Z basis.)
- Show that the first Bell state can also be written as $\frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$. Hence if both qubits are measured in the Z basis the results always agree, and if both are measured in the X basis the results again always agree.
- Show that no product state $|\alpha\rangle \otimes |\beta\rangle$ can give perfectly agreeing results in *both* bases. *Hint: agreeing with certainty in the Z basis forces each of $|\alpha\rangle, |\beta\rangle$ to be $|0\rangle$ or $|1\rangle$. What do these give in the X basis?*

7. GHZ states: fan-out versus chain. In Worksheet 1 you met the fan-out circuit H on qubit 0 followed by $\text{cx}(0, 1)$, $\text{cx}(0, 2)$, $\text{cx}(0, 3)$. A cheaper-looking alternative is the chain: H on qubit 0, then $\text{cx}(0, 1)$, $\text{cx}(1, 2)$, $\text{cx}(2, 3)$.

- Show that on the input $|0000\rangle$ both circuits produce the four-qubit GHZ state $\frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$.
- Show that the two circuits are nevertheless *different* unitaries by finding a computational basis input on which their outputs differ. (Moral: agreeing on one input does not make two circuits equal.)
- Qubit 0 of the GHZ state is measured in the Z basis. Give the outcome probabilities and the state of the remaining three qubits after the measurement. Is that state entangled?

- (d) Now qubit 0 is measured in the X basis instead. *Hint: rewrite $|0\rangle$ and $|1\rangle$ in terms of $|\pm\rangle$.* Give the outcome probabilities and the state of the remaining three qubits for each outcome. Is that state entangled? Comment on the difference from (c).

8. On a computer (optional).

In Qiskit, build the two circuits of Q3 and compare `Operator(qc1).equiv(Operator(qc2))`. Then do the same for Q4(a) against `qc.swap(0,1)`, and for the two circuits of Q7(b). Remember the footnote on bit ordering when you print states.